Available online at www.sciencedirect.com
ScienceDirect
Optics & Laser Technology 39 (2007) 1461–1471
Optics & Laser Technology
www.elsevier.com/locate/optlastec

Study of a line width estimation model for laser micro material processing using a photodiode

Young Whan Park[a], Sehun Rhee[b,*]

[a]Department of Mechanical Engineering, Hanyang University, 17, Haengdang-dong, Sungdong-gu, Seoul 133-791, Republic of Korea
[b]School of Mechanical Engineering, Hanyang University, 17, Haengdang-dong, Sungdong-gu, Seoul 133-791, Republic of Korea

Received 4 January 2006; received in revised form 28 September 2006; accepted 28 September 2006
Available online 22 November 2006

Abstract

Estimation of the line width for a laser marking on the silicon wafer is very important to improve the productivity of the final product which use nonpackaged chip. Until now, only theoretical and numerical estimation models have been studied. However, it is not easy for these models to apply to real systems. In this study, a process monitoring system was used to develop an estimation model for the laser marking width. The plasma produced by interaction between the laser and the wafer was measured using an optical sensor. For each laser power setting, the correlation between the signal acquired from the optical sensor and the resulting line width was analyzed. Estimation models were developed for laser marking width through statistical regression analysis and an artificial neural network algorithm.

Keywords: Laser marking; Line width estimation; Estimation models

1. Introduction

Semiconductor industry is one of the most important electronics industries. All components, including semiconductor circuits, are being reduced in size. As semiconductor chips become smaller, they are becoming more integrated. Semiconductor chips fall into two basic categories: packaged and nonpackaged chips. Packaged chips are encased in a plastic molding on which product information such as the manufacturer's name, model number, bar code and logo are located. However, this information is placed directly on the back of the chip for nonpackaged chips. Producing nonpackaged chips for industrial use requires a system for marking the product information and for estimating the quality of the marking.

There are several marking technologies currently in use. Laser marking has advantages over the other marking processes, including that it is a precise and high-speed process with excellent reproducibility and flexibility. Laser marking at the wafer level is performed before dicing, which is the final processing step to make a chip. Therefore, it is very important to increase the marking quality. The criterion of laser marking quality is readability, which is closely related to marked region characteristics, such as ablation of depth and line width. In order to estimate the marking quality in real time, it is necessary to know the line width during the marking process.

The geometry of the marked region is dependent on the heat transfer and plasma phenomena when the material is irradiated by the laser. Chan and Mazumder [1] estimated the material removal rate from the vaporization and liquid material ejection using a 1-D model. Kar and Mazumder [2,3] estimated the melting depth with respect to the effect of multiple reflections in the cavity using a 2-D model. Modest [4] estimated the groove shape using a 3-D model and compared the ablation processes of a CW laser, pulse laser, and Q-switched laser. In addition, he estimated the geometry using a numerical model as well as actual experiments. Peligrad [5,6] developed a melting depth model for the laser marking and engraving of clay tile, and verified this model through experiments. He also obtained the laser marking width according to the laser

*Corresponding author. Tel.: +82 2 2220 0438; fax: +82 2 2299 6039.
E-mail addresses: honeydew@hanyang.ac.kr (Y.W. Park), srhee@hanyang.ac.kr (S. Rhee).

0030-3992/$ - see front matter
doi:10.1016/j.optlastec.2006.09.013

power and the travel speed using the ARX model. He used a PI controller to control the laser power. Zhou [7] suggested a theoretical model for estimating the cutting depth of nonmetallic materials such as acrylic plastic Polymethylmethacrylate (PMMA), and modified the theoretical formula by comparing the results from the model with experimental results. Miyamoto [8] compared the theoretical drilling rate and actual drilling rate from the laser heat input into the wafer during drilling using a KrF excimer UV Laser. Gu [9] investigated the effect of laser power, laser repetition rate, marking speed, and pulse overlap on the marking of the wafer.

The studies mentioned above have contributed to the theoretical understanding of laser marking but application of their results to real systems is limited. Studies focused on process monitoring methods have more practical applications. These methods have been used for laser material processing such as laser welding and cutting, and the estimation of material processing geometry. Acoustic emission, audible sound, infrared sensing, and ultraviolet sensing are used as measurement tools in laser welding [10]. For optical measurements, photodiodes are most widely used. Tu [11,12] and Rhee [13] estimated the shapes of welding beads using an optical sensor.

In this paper, a monitoring system that measures the plasma produced during laser marking was built and marking width estimation models using this monitoring system were developed. Because laser marking is a high-speed process, the plasma produced during the process was measured in real time by constructing a coaxial monitoring system to synchronize the marking point and the measurement. Statistical regression models and an artificial neural network model were used to estimate the width of laser marking based on the relationship among the laser power, the line width and the measured signals from the sensor.

2. Experiments

2.1. Experimental setup

The laser marking monitoring tests were performed using a 3 W-Class diode-pumped solid-state (DPSS) green laser with a wavelength of 532 nm. The cross sectional distribution of the laser beam was TEM_{00} mode. A Q-switched pulse-type oscillator with a repetition rate range of 1–100 kHz was used. The nominal pulse width of the laser was 70 ns at 7.5 kHz. The laser configuration used is shown in Fig. 1.

The laser beam coming from the generator was magnified through a beam expander and irradiated to the designated position by the galvano-scanner. The laser was then focused on the silicon wafer by the flat-field focus lens as shown in Fig. 1. The marking was performed on the surface of a silicon wafer.

Because the laser marking was carried out at a high speed by the galvano-scanner, it was impossible to monitor the laser marking using an external sensor. If a sensor was

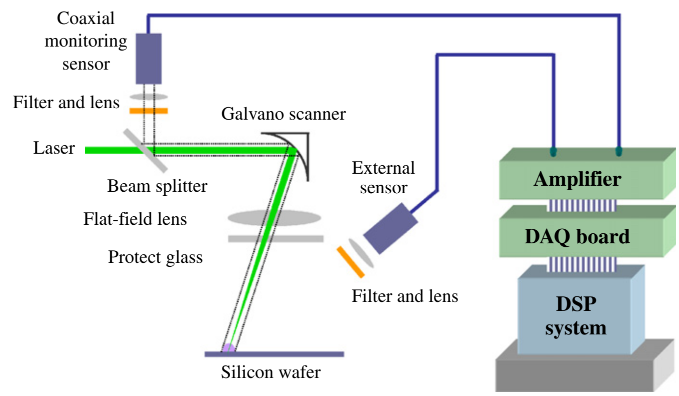

Fig. 1. Schematic of laser-marking monitoring system.

installed behind galvano-scanner, the marking area could be monitored in real time. However, since the light from the plasma and the laser beam were on the same axis, additional optical components such as a splitter, a filter and an internal sensor mount, were needed to be installed. This measuring system is, therefore, referred to as a coaxial monitoring system. In order to monitor the laser marking process with the coaxial monitoring system, a beam splitter was installed at an angle of 135° to the path of laser and the coaxial monitoring sensor was installed behind the beam splitter as shown in Fig. 1. The sensor for monitoring laser marking was a Si photodiode, which has a response range of 450–1050 nm and a peak wavelength of 900 nm. A long pass filter was used to remove the laser wavelength of 532 nm. It was installed to receive the beam resulting from interaction between the laser and the silicon wafer. The filter was needed because the measuring sensor had a wide response range. Also, a lens with a 50 mm focal length was installed in front of the sensor in order to effectively measure the plasma light generated during the laser marking. The sampling rate of data acquisition was 200,000 samples per second.

2.2. Experimental conditions

There are many parameters to control for laser marking, including the laser power, the laser wavelength, the marking speed, and the focal position. In this experiment, the relationship between the control parameters and the marking line width was studied, and the correlation between the marking quality and the sensor signals was proposed.

In this study, the laser power was used as a main control parameter for the laser marking experiment. Laser power ranged from 1 to 3 W and was divided into 9 levels by 0.25 W increments. The pulse repetition rate was set to 10 kHz. Each test was repeated 6 times and 54 tests were performed.

Readability on a wafer is closely dependent on the line width. Width is more important than the depth of marking in cognition [9] because sufficient width is required in order to recognize the marked section. Therefore, line width was used to assess laser-marking quality. Fig. 2 shows an example of the marked results. The relationship between the line width, acquired from 9 lines and the signals acquired from the photodiode was analyzed.

3. Results and discussions

Fig. 3 shows photographs of marked shapes produced at different laser powers. Fig. 4 shows that the line width increases as the laser power increases. Experiments were performed at a constant marking speed, so the marking width was determined only by the laser power.

When the readability as a function of the width was considered, most of the lines were recognizable under all illumination conditions when the width was greater than 40 μm. The lines were not easy to read when the width was 30–40 μm. When the width was less than 30 μm, an optical microscope was required to observe the lines because the marked region could not be visually verified. Thus, the laser output should be greater than 1.5 W, which corresponds to a line width of approximately 40 μm.

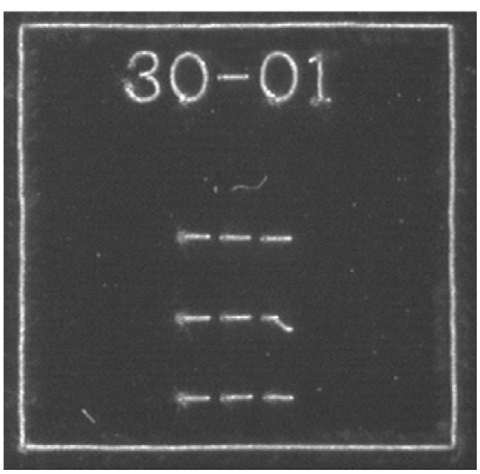

Fig. 2. Example of laser marking used in experiments.

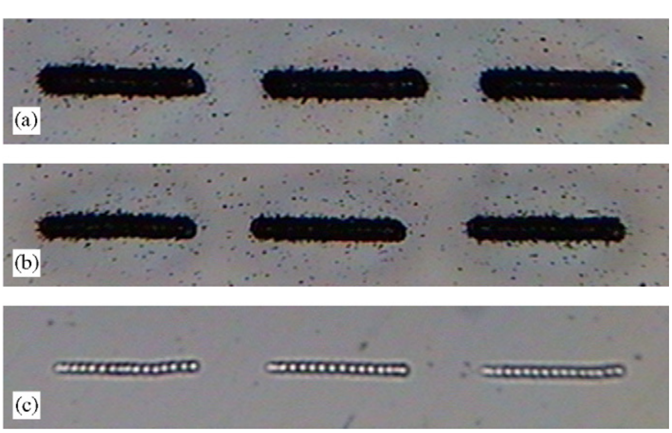

Fig. 3. Marking shapes for different laser powers: (a) 3.0 W; (b) 2.0 W and (c) 1.0 W.

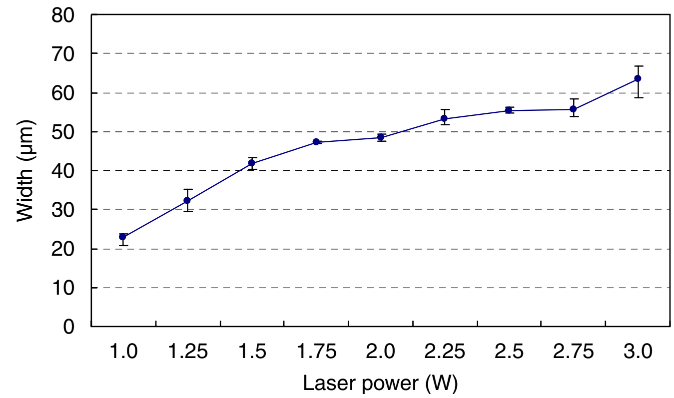

Fig. 4. Marking width according to laser power.

Fig. 5 shows the signal from the photodiode, which measured the plasma produced during laser marking. As can be seen here, whenever a marking point was produced on the wafer surface by one pulse of the laser, one peak of signal was obtained. The marked line consists of many marked points produced by pulses of the laser. In Fig. 5(d), a large-scale graph of sensor signals is also shown, and a signal peak is repeated every 0.1 ms because the laser repetition rate was 10 kHz. The sensor signals before and after the peak are symmetric, and therefore the measured signal can be regarded as reliable.

As shown in Fig. 5, the sensor signal varied according to the laser power. From the sensor signals, three derived variables were obtained: the average peak value, the average of integrated value for each pulse signal, and the ratio of standard deviation of the peak value to the average peak value. These values are shown in Figs. 6–8, respectively. Figs. 6 and 7 show the average peak value and the average of integrated value for each pulse signal for various laser powers. As shown in these figures, the average peak value and the average of integrated value for each pulse signal were directly proportional to laser power. The interaction between the laser and silicon wafer increased as the power was increased, and the intensity of the optical signal increased accordingly. Thus, as the laser power was increased, the peak value was increased and the total amount of plasma light produced during irradiation also increased. Fig. 8 shows the ratio of standard deviation of the peak value to the average peak value according to laser power. This parameter was inversely proportional to laser power. When the laser power was high, such as 2.0–3.0 W, laser ablation due to the interaction between the laser and silicon wafer occurred readily and the plasma produced during the ablation was stable. Therefore, the peaks of the sensor signal were stable and standard deviations of the peaks were low. When the power was low, however, the measured signal had a relatively increased standard deviation of the peak value because of instable interaction,

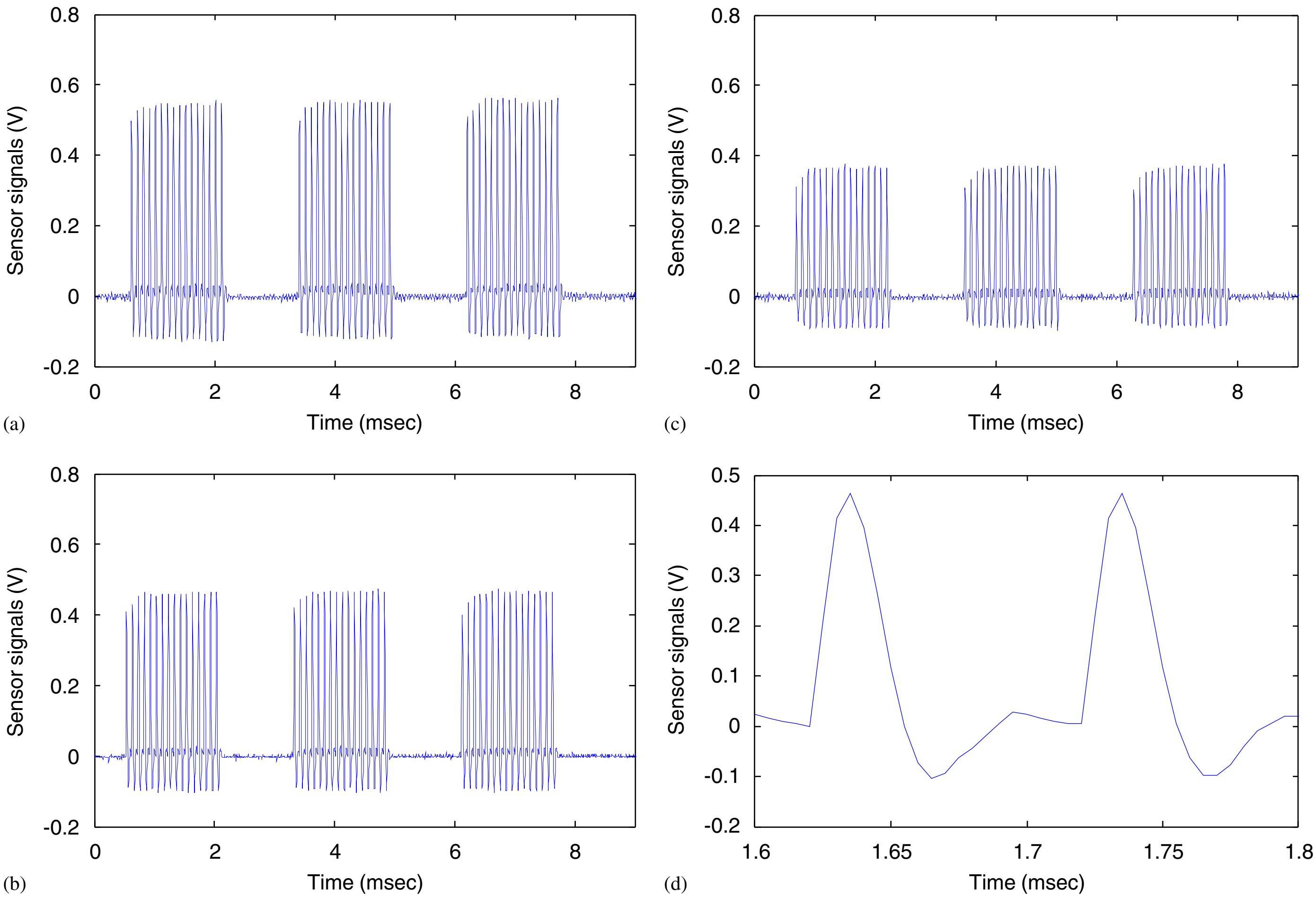

Fig. 5. Signals from photodiode for different laser powers: (a) 3.0 W; (b) 2.0 W; (c) 1.0 W and (d) large scale of sensor signal at 2.0 W.

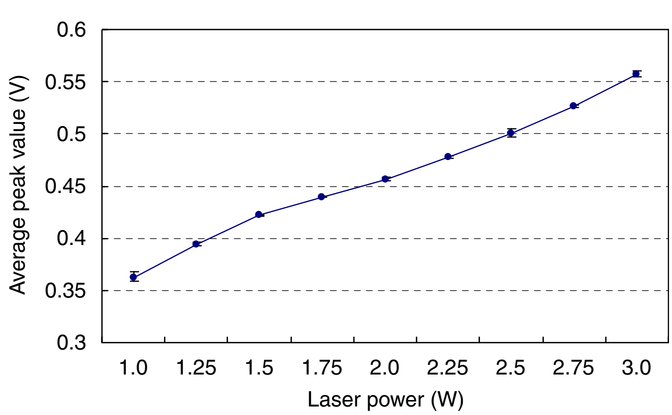

Fig. 6. Average of peak values in sensor signals according to laser power.

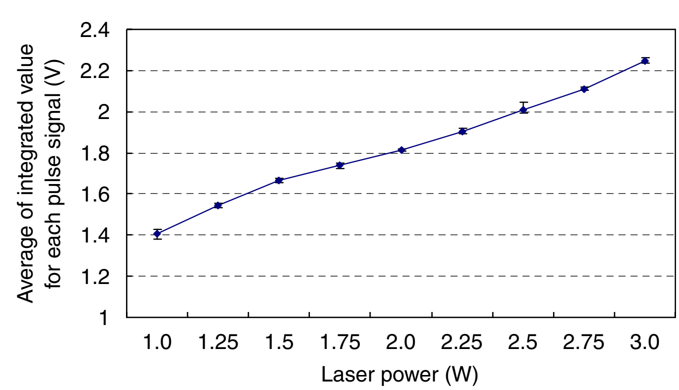

Fig. 7. Average of integrated value for each pulse signal according to laser power.

so the value of the ratio of standard deviation of the peak value to the average peak value was higher.

According to the experimental results, the marking width and sensor signal were affected by the laser power. Figs. 4, 6–8 show that when the laser power was changed, the phase of the base material and ablation mechanism were changed by the interaction between the laser and wafer. The optical signal and marking width were also changed accordingly. As a result, there is a relationship between line widths and sensor signals according to the laser power. Therefore, the model could be configured to estimate the line width using the sensor signals.

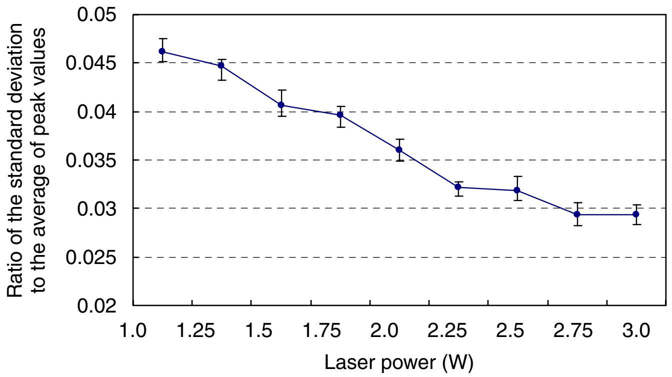

Fig. 8. Ratios of standard deviations to average peak values according to laser power.

Table 1
Preset values for normalizing feature values

	Average of peak value	Average of integrated value for each pulse	Ratios of standard deviation to average of peak value
x_{max}	0.6112	2.426	0.055
x_{min}	0.2912	1.0551	0.02

Table 2
Coefficients for multiple linear regression model

β_0	β_1	β_2	β_3
26.01	6.34	43.05	−12.23

4. Width estimation model

4.1. Feature extraction

Statistical regression models and an artificial neural network model were established to estimate the marked line width based on the experimental results described in the previous section. Measurements should be extracted from the sensor signals before estimating the marked line width. It was shown that the characteristic factors of the signals are closely related to the marking width according to the laser power.

The average peak value, the average of integrated value for each pulse signal and the ratio of standard deviation of the peak value to the average peak value were used. Because sensor signals give us more information for laser marking than laser power, these values extracted from the raw sensor signals were used as the input parameters of the estimation model (as the input parameters to the estimation models and they were extracted from raw data.) The significance of each input parameter against the output parameter varies because the range of feature values is different as shown in Figs. 6–8. Each characteristic factor had a different range and hence each should be normalized. The normalizing formula is

$$x_{\text{normalized}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (1)$$

where $x_{\text{normalized}}$ is the normalized characteristic factor and x is the value of the characteristic factor acquired from the sensor signal. $x_{\min}$ and $x_{\max}$ are preset values. When the laser power was 0.5 W, very fine marking was produced. However, if the laser power was less than 0.5 W, no marking was performed. Therefore, $x_{\min}$ was defined as the average peak value, the average of integrated value for each pulse signal and the ratio of standard deviation of the peak value to the average peak value when the laser power was 0.5 W. If the laser power was greater than 3.4 W, the marking process damaged the electric circuit of the chip. Therefore, the $x_{\max}$ was set as the feature value when the

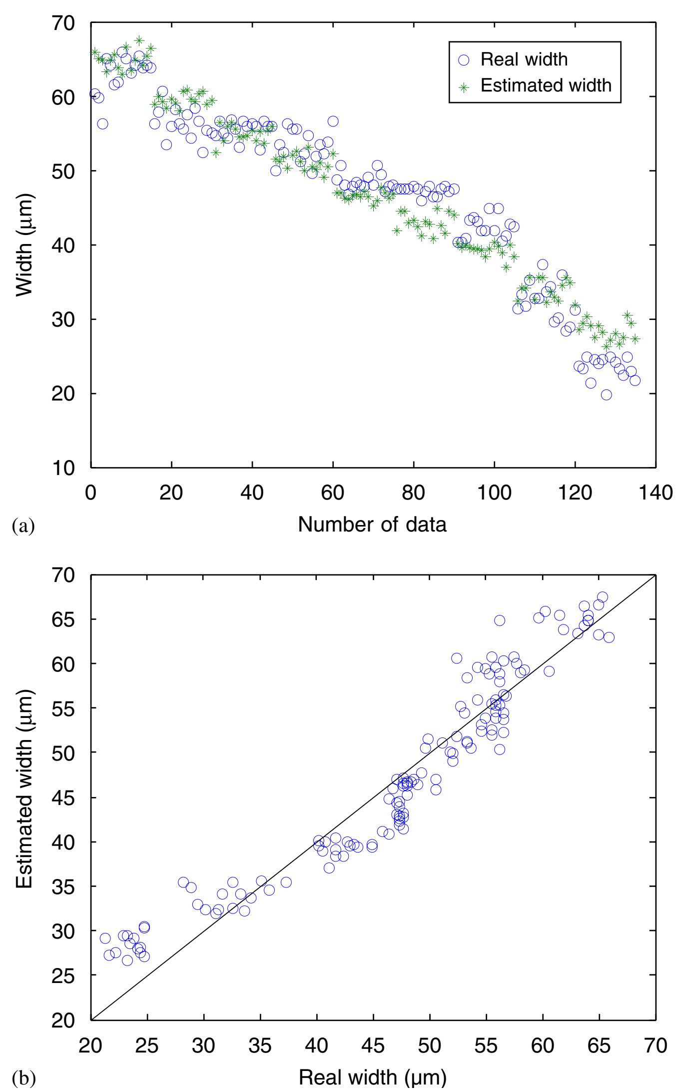

Fig. 9. Results of multiple linear regression model using regression data: (a) marking width estimation and (b) comparison between real value and estimated value.

laser power was 3.4 W, which was the maximum laser output. The values of $x_{\min}$ and $x_{\max}$ for each feature are shown in Table 1.

4.2. Multiple linear regression model

The line width estimation model was developed using the normalized features of sensor signals described in Section 4.1. For each 9 levels of laser power, the experiments were repeated 6 times. Five of the 6 sets of experimental data were used to define the estimation model and the other data set was used to verify the estimation model. Nine lines were obtained for each experiment as shown in Fig. 2. The average value for three lines in the same row was taken as one data point; so, three data sets were obtained for each test. The total number of data

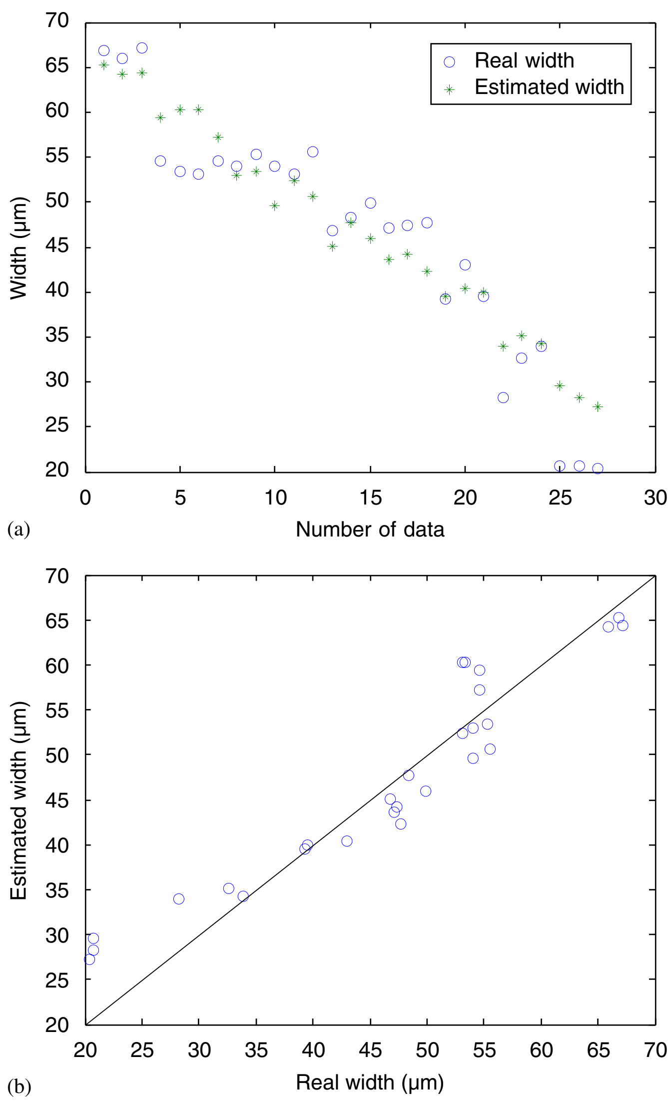

Fig. 10. Results of multiple linear regression model using evaluation data: (a) marking width estimation and (b) comparison between real value and estimated value.

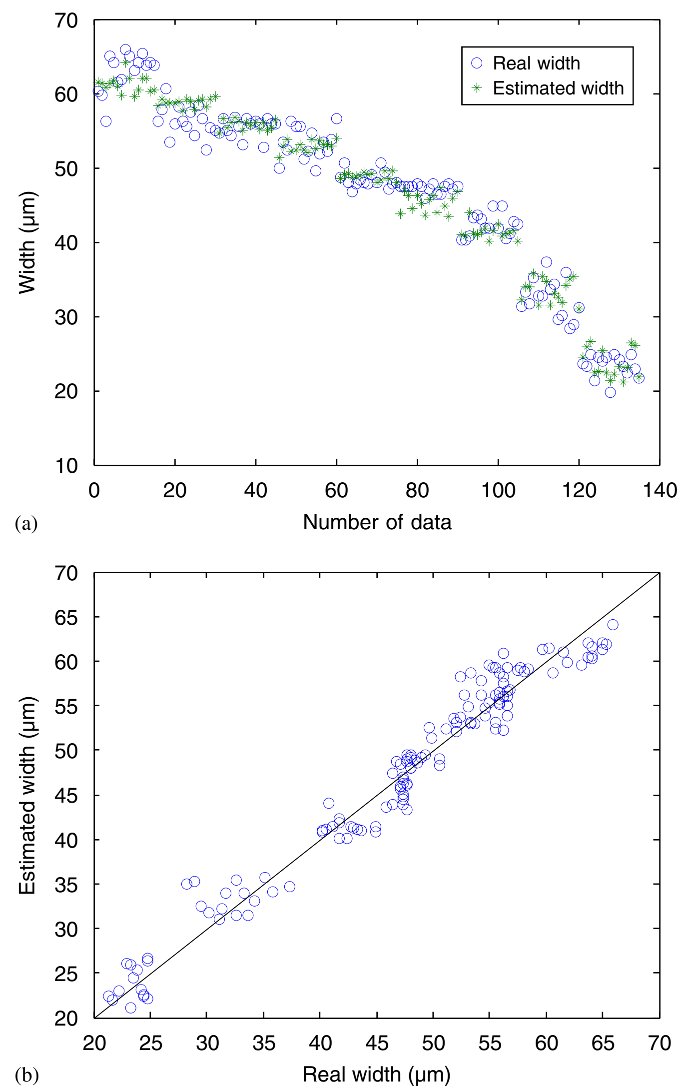

Fig. 11. Results of second-order polynomial regression model using regression data: (a) marking width estimation and (b) comparison between real value and estimated value.

Table 3
Coefficients for second-order polynomial regression model

β_0	β_1	β_2	β_3	β_4	β_5	β_6	β_7	β_8	β_9
−15.41	255.42	−108.45	29.89	−2892.31	−284.41	227.48	1229.85	1596.61	−20.01

points to define the estimation model was 135, and 27 data points were left to verify this model.

The normalized values of the average of integrated value for each pulse signal, the average peak value, and the ratio of standard deviation of the peak value to the average peak value were set as the input parameters of the multiple linear regression model. These three values were defined as x_1, x_2, and x_3, respectively. The marked line width was set as the response variable and defined as $\hat{y}$.

The multiple linear regression model can be expressed as

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \tag{2}$$

where β_0, β_1, β_2, and β_3 are coefficients. These coefficients were acquired by the least-squares method. The values of each coefficient are shown in Table 2. Because all input variables were normalized, these coefficients represent the influence of each input variable on the output variable. As shown in Table 2, the average peak value (x_2) plays the most important role in the multiple linear regression model because β_2 is greater than any other coefficient.

The estimation performance of the proposed regression model is shown in Fig. 9. This regression model estimated a little lower than the measured value in the middle part, and higher than the real value at the end part of the data because a first-order regression model was used. Its estimation performance was limited for some regions with

Table 4
Coefficients for multiple nonlinear regression model

β_0	β_1	β_2	β_3
67.61	−2.725	31.83	−0.17

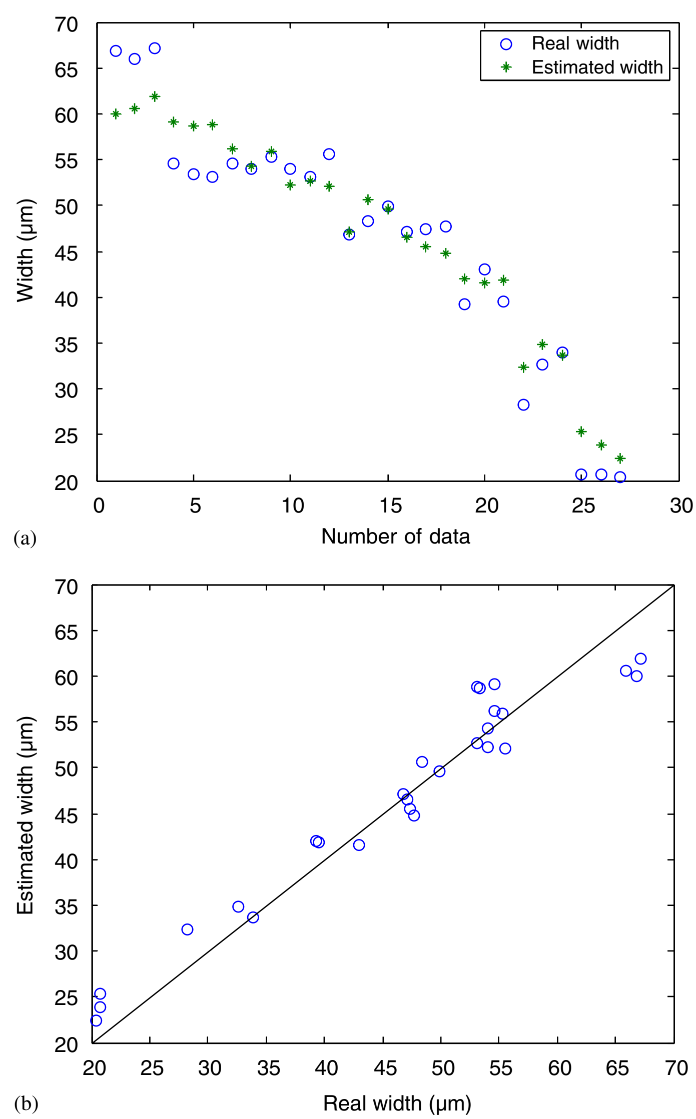

Fig. 12. Results of second-order polynomial regression model using evaluation data: (a) marking width estimation and (b) comparison between real value and estimated value.

a nonlinear tendency. Fig. 10 shows the estimation performance using the evaluation data set, which was not used to define the regression model.

4.3. Second-order polynomial regression model

The second-order polynomial regression model was developed in the same manner as the multiple linear regression model. This regression model is defined as

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_2 + \beta_6 x_2 x_3 + \beta_7 x_1^2 \beta_8 x_2^2 + \beta_8 x_3^2. \quad (3)$$

The values of each coefficient are shown in Table 3. It is clear that the multiplication of the average of sum of signal values per pulse with the average peak value ($x_1 x_2$), the square of the average of sum of signal values per pulse (x_1^2) and the square of the average peak value (x_2^2) had the most effect on the output variable in this model.

Fig. 11 shows the comparison between the estimated line width using the second-order polynomial regression model and the measured line width. The estimation performance of the second-order regression model was better than the linear model because it includes second-order terms. Fig. 12 shows the estimation performance of the second-order polynomial regression model using the verification data set that was not used to define this regression model.

4.4. Multiple nonlinear regression model

The multiple nonlinear regression model was formulated as

$$\hat{y} = \beta_0 + \beta_1 \ln x_1 + \beta_2 \ln x_2 + \beta_3 \ln x_3. \quad (4)$$

The values of each coefficient are shown in Table 4. The logarithm of the average peak values ($\ln x_2$) greatly influenced the output variable.

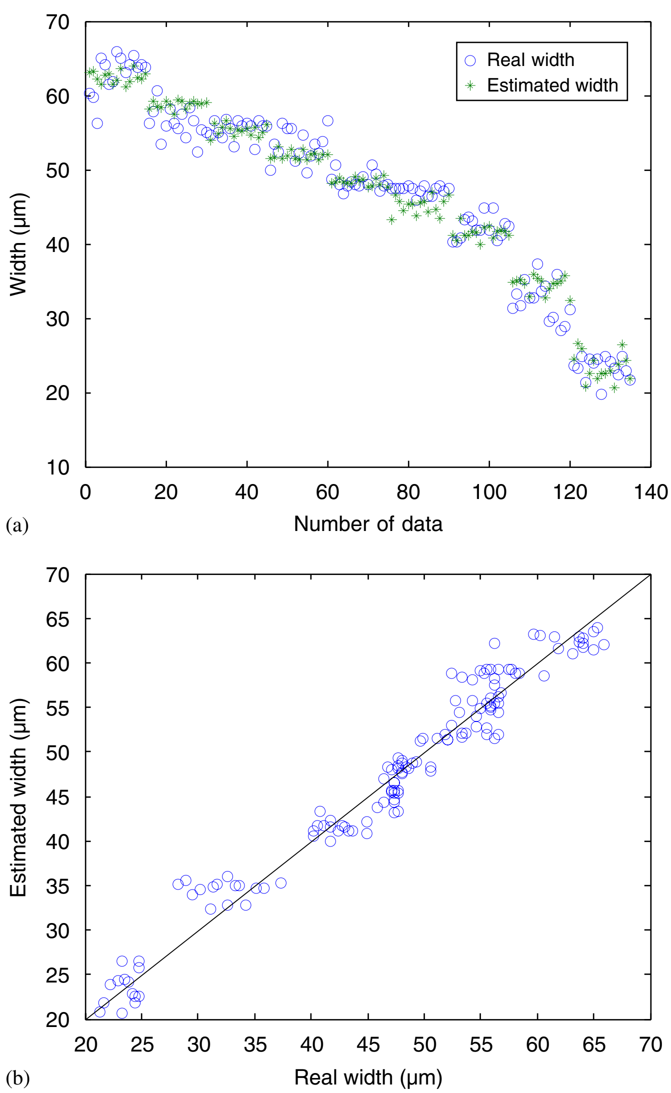

Fig. 13. Results of multiple nonlinear regression model using regression data: (a) marking width estimation and (b) comparison between real value and estimated value.

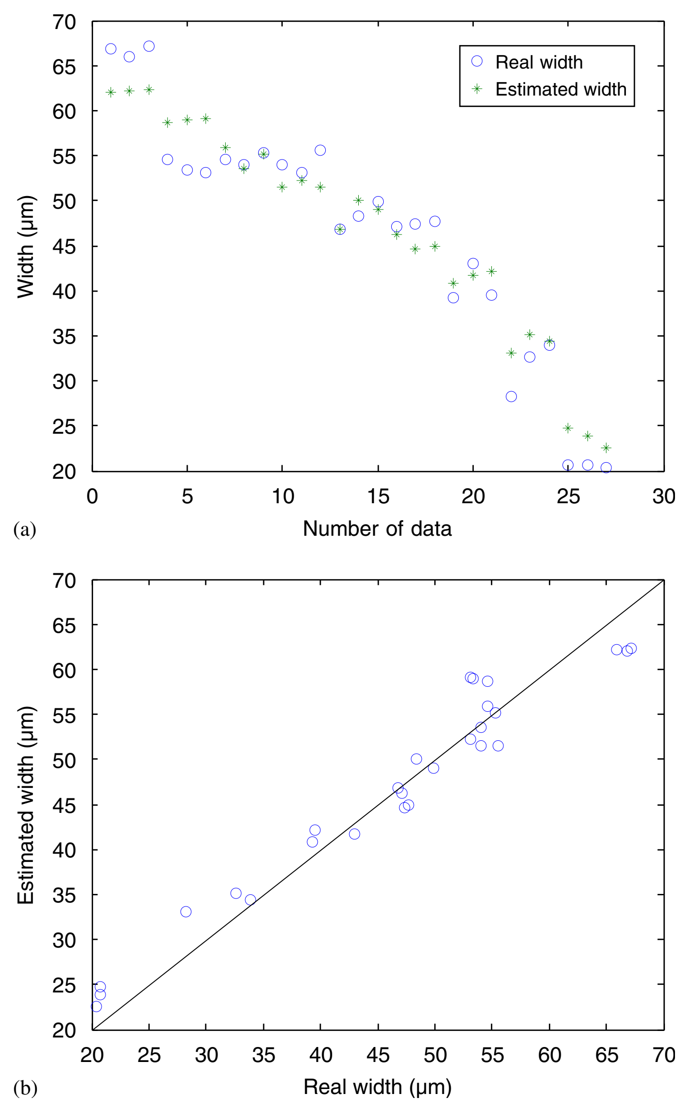

Fig. 14. Results of multiple nonlinear regression model using evaluation data: (a) marking width estimation and (b) comparison between real value and estimated value.

Fig. 13 shows the estimation performance for the multiple nonlinear regression model. Fig. 14 shows the estimation performance for the evaluation data.

4.5. Artificial neural network model

The neural network model, one of the artificial intelligence algorithms, was also used to estimate the width of the marking line by measuring the optical signal produced during laser marking. The neural network model was trained using 135 data points and verified using 27 data points in a manner similar to that described in the previous section.

For the configured neural network model, the normalized values of the average of sum of signal values per pulse, the average peak values and the ratio of standard deviation of the peak values to the average peak values were set as the input nodes, while the line width of laser marking was set as the output node. Two hidden layers were configured to have 7 nodes and 5 nodes, respectively. The form of the configured layer is shown in Fig. 15. The back propagation algorithm was used as a learning method. The learning rate was set to 0.1 and the momentum parameter was 0.9. The minimum learning error was less than 1 and maximum learning repetition number was 100,000 times.

Fig. 16 shows the comparison between the laser marking width used to teach the artificial neural network model and the estimated marking width output by the neural network model. The artificial neural network model enabled us to learn the input and output factors up to the set learning error, so it enabled us to acquire a more reliable estimation than the regression models. As shown in Fig. 16, the neural network model has excellent estimation performance. In Fig. 17, the estimation performance using new data which had not been used for learning in the neural network model was obtained.

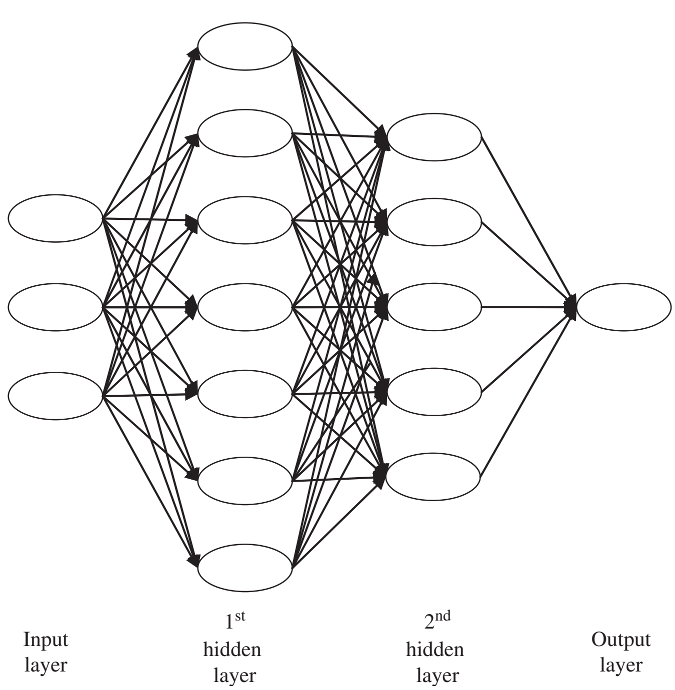

Fig. 15. Configured neural network model for width estimation.

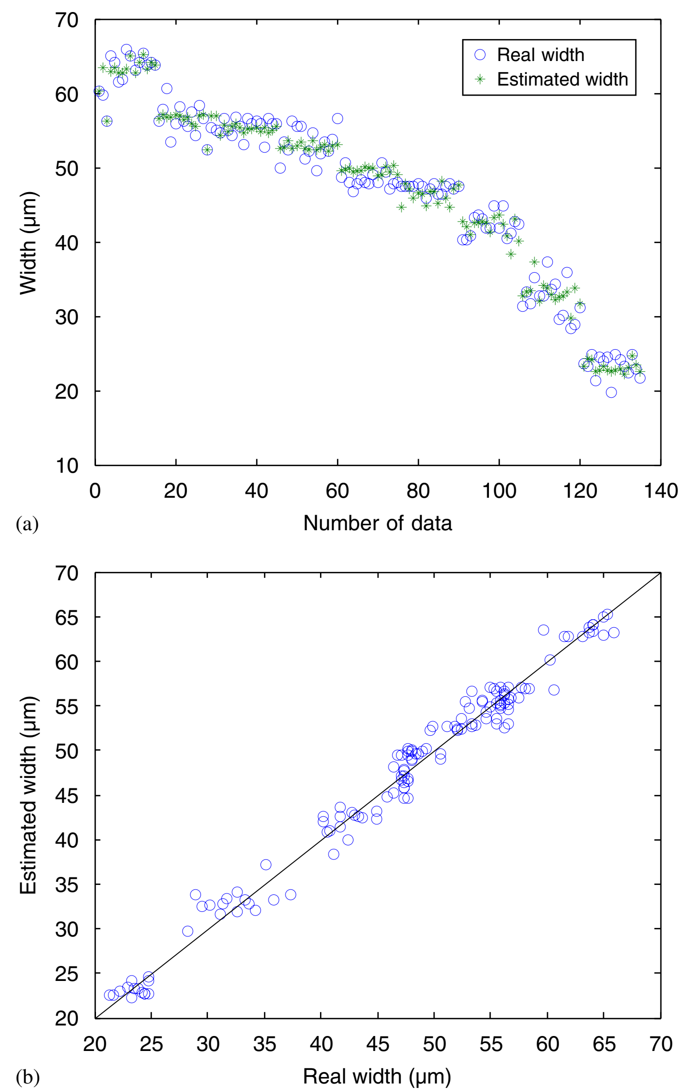

Fig. 16. Results of neural network model using training data: (a) marking width estimation and (b) comparison between real value and estimated value.

4.6. Performance comparison of the four models

Generally, the *F*-test by analysis of variance, coefficient of determination and residual average square are used to indicate the accuracy of a regression model [14]. However, the *F*-test and residual average square were not acquirable in the neural network model. The estimation performance of a model was therefore determined in two ways: the coefficient of determination and average error rate (AER) for the four estimation models. The formula used to acquire the coefficient of determination is shown in Eq. (5). AER is defined in Eq. (6). When the coefficient of determination is larger or the AER is smaller, the

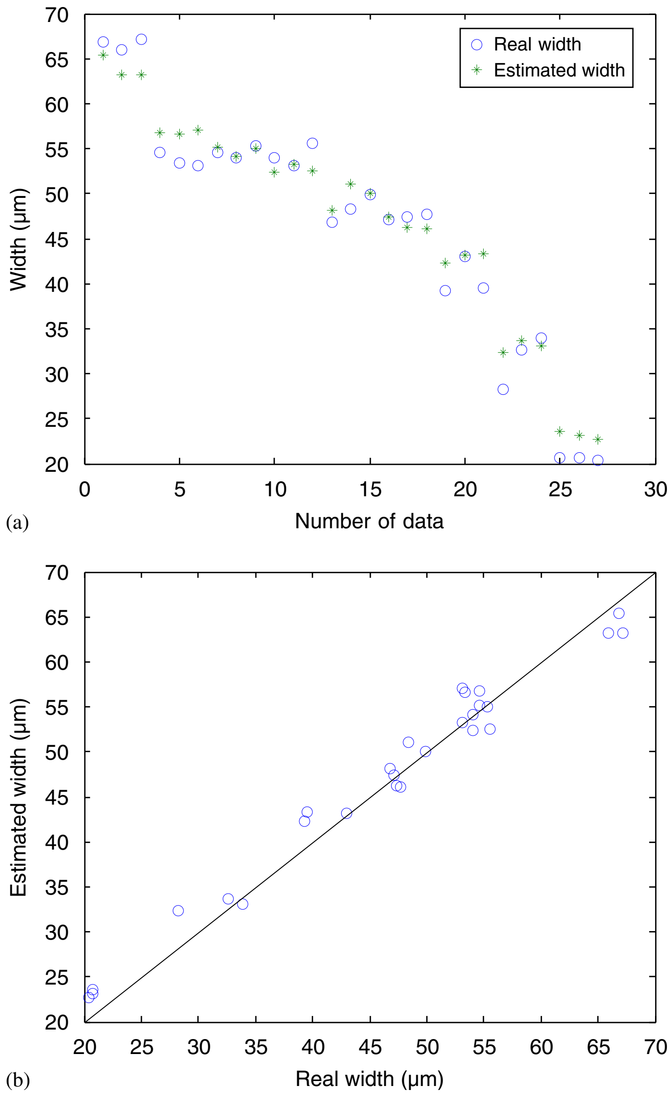

Fig. 17. Results of neural network model using evaluation data: (a) marking width estimation and (b) comparison between real value and estimated value.

estimation performance becomes better:

$$R^2 = \frac{\sum_{i=1}^{n}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sum_{i=1}^{n}(y_i - \bar{y})^2 \sum_{i=1}^{n}(\hat{y}_i - \bar{\hat{y}})^2}, \tag{5}$$

$$\mathrm{AER} = \frac{1}{n}\sum_{i=1}^{n}\frac{|y_i - \hat{y}_i|}{y_i}. \tag{6}$$

In these equations, R^2 indicates the coefficient of determination, n indicates the number of experimental data or estimated data and i is the data number. y is the measured line width and $\hat{y}$ indicates the estimated width by each model. $\bar{y}$ and $\bar{\hat{y}}$ are the average of measured values and estimated values, respectively.

Table 5 shows the coefficients of determination for each estimation model with the regression and the evaluation data. As shown in Table 5, the estimation performance of the neural network model was better than any other model, based on the coefficients of determination. Table 6 shows the AER of the regression and training data of the evaluation data. As shown in Table 6, the neural network model has the best estimation performance of all models, based on the average error rate. As shown in Fig. 4, the marking width increased somewhat nonlinearly as the laser power increased. In Figs. 6–8, the feature values of the measured signal are linearly related to the laser power. Therefore, the linear models had a limitation in estimating the marking width. The nonlinear models were more accurate than the linear models. Tables 5 and 6 show that the estimation performance of the neural network model was twice as accurate as the multiple linear regression model. Moreover, the estimation performance of the neural network model was 1.3 times more accurate than the second-order polynomial regression model and the nonlinear regression model.

Table 5
Coefficient of determination for each estimation model

	Multiple linear regression model	Second-order polynomial regression model	Multiple nonlinear regression model	Neural network model
Regression/training data	0.914	0.963	0.96	0.981
Evaluation data	0.903	0.946	0.952	0.977

Table 6
Average error rate for each estimation model

	Multiple linear regression model	Second-order polynomial regression model	Multiple nonlinear regression model	Neural network model
Regression/training data	0.073	0.042	0.044	0.031
Evaluation data	0.097	0.064	0.063	0.048

5. Conclusion

Real-time monitoring of the quality during laser marking is a very important issue for the maximization of productivity and efficiency. In order to monitor the laser marking, a photodiode was used to measure the plasma light emitted during laser marking. The photodiode on a coaxial line was able to accurately record light emission during the high speed of the marking process. Also, the relationship between the marking width and sensor signal as a function of the marking condition was obtained. The line width increased as the laser power was increased, and the interaction between the laser and the wafer also increased accordingly.

Using the experimental results, four statistical regression models were developed to estimate the line width that represented the marking quality. A multiple linear regression model, second-order polynomial regression model, multiple nonlinear regression model and neural network model were used to estimate the line width. The coefficient of determination and average error rate of each model were calculated in order to quantitatively compare the accuracy of each model. It was shown that the neural network model was better than any other model in terms of both the coefficient of determination and the average error rate in the estimation performance.

Acknowledgments

This work was supported by the Second Brain Korea 21 Project in 2006.

References

[1] Chan CL, Mazumder J. One-dimensional steady-state model for damage by vaporization and liquid expulsion due to laser-material interaction. J Appl Phys 1987;11:4579–86.
[2] Kar A, Mazumder J. Two-dimensional model for material damage due to melding and vaporization during laser irradiation. J Appl Phys 1990;8:3884–91.
[3] Kar A, Rockstroh T, Mazumder J. Two-dimensional model for laser-induced materials damage: effects of assist gas and multiple reflections inside the cavity. J Appl Phys 1992;6:2560–9.
[4] Modest MF. Three-dimensional, transient model for laser machining of ablating/decomposition materials. Int J Heat Mass Transfer 1996;2:221–324.
[5] Peligrad A, Zhou E, Morton D, Li L. A melt depth prediction model for quality control of laser surface glazing of inhomogeneous materials. Opt & Laser Technol 2001;1:7–13.
[6] Peligrad A, Zhou E, Morton D, Li L. Dynamic models relating processing parameters and melt track width during laser marking of clay tiles. Opt & Laser Technol 2002;2:115–23.
[7] Zhou BH, Mahdavian SM. Experimental and theoretical analyses of cutting nonmetallic materials by low power CO_2-laser. J Mater Process Technol 2004;2:188–92.
[8] Miyamoto I, Asada S, Sano T, Ohmura E. High speed drilling of thin silicon wafer by UV laser. Pro ICALEO 2001: M602.
[9] Gu B. Laser wafer marking at die level. Proc ICALEO 2001:M408.
[10] Sun A, Kannatey-Asibu E, Gartner M. Sensor systems for real-time monitoring of laser weld quality. J Laser Appl 1999;4:153–68.
[11] Tu JF, Lankalapalli K, Leong KH, Gartner M. On-line estimation of laser weld penetration. Trans ASME J Dyn Syst 1997;119:791–801.
[12] Lankalapalli K, Tu JF, Gartner M. A model for estimating penetration depth of laser welding process. J Phys D: Appl Phys 1997;29:1831–41.
[13] Park H, Rhee S. Estimation of weld bead size in CO_2 laser welding by using multiple regression and neural network. J Laser Appl 1999;3:143–50.
[14] Montgomery DC, Peck EA. Introduction to linear regression analysis. 2nd ed. New York: Wiley; 1992.